

Section 2

Digital Transformation across Industries

Chapter 7

Digital in Life Sciences

- Life sciences has been one of the laggards but is now at a tipping point for digital transformation.
- To win, players must consider a fundamental shift in the business model by putting the patient at the centre. Historically, it is the clinician that has been the customer in this industry.
- Four imperatives to drive successful execution in this industry are: cloud, automation, a partnering ecosystem and driving open innovation.

Key Takeaways

Reimagining innovation in R & D

Rise of Digital Patient Engagement Models

Lesson 1: Innovation and customer-centricity

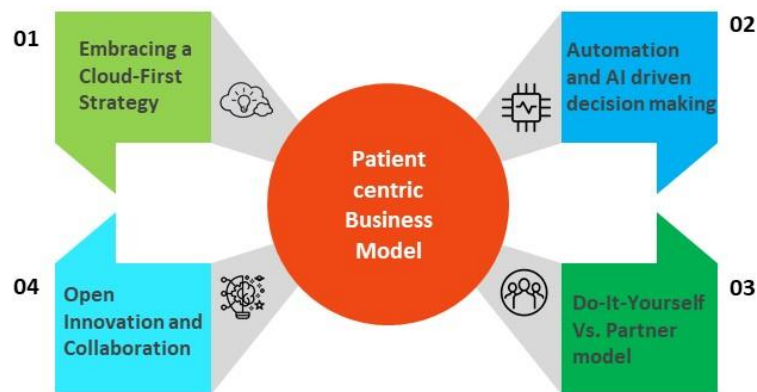
Nitin's introduction:

'What business are you in?' – This question, raised by Professor Theodore Levitt in a Harvard Business Review (2006) article, is a fundamental one that life sciences manufacturers are faced with. It lies at the heart of the digital transformation taking place in this industry. Pharmaceutical manufacturers have traditionally built businesses focused on product superiority, targeting clinicians (their 'customer') who have patients (the end consumers of their products). With digital becoming all-pervasive and with the industry moving towards specialty and personalized treatments, understanding the needs of the end consumer has never been more essential, in the process leading to a redefinition of who their end customer should be – The Patient!

New and non-traditional entrants in the life sciences industry, many of whom are digital-native companies, add another dimension of disruptive forces at work, driving the need for digital transformation to be the centerpiece of a pharmaceutical company's future-state strategy. The pharmaceutical industry, unlike other sectors, has historically been a laggard in digital adoption. A 2018 Deloitte-MIT Sloan Review study found that less than 20 per cent of pharmaceutical companies had currently taken decisive and bold moves to take advantage of digital capabilities. Together, these developments present a tremendous opportunity for forward-looking

companies to shape their digital agenda to address their primary concern of serving patients better.

Life Sciences - Shifting Paradigm from Clinician to Patient Digital Fundas A CSR initiative of mncs



Proprietary and confidential

Digital presents new ways to Life Sciences companies to provide value to customers (patients) and create a competitive advantage for themselves. Harnessing data and leveraging cloud-based technologies, combined with a deep understanding of customers and their unmet needs, offer new possibilities for innovation in R&D and delivery of differentiated multi-channel patient engagement programmes for improving patient outcomes.

Reimagining innovation in R&D

Linear methods of clinical research and discovery for new therapies face the continuing pressure of increasing R&D costs and shorter product ('medicine') life cycles.

While placebo-controlled clinical trials remain the gold standard, synthetic control arms offer new ways of assessing new treatment interventions. The US Food and Drug Administration's (FDA) approval of Brineura from Biogen for Batten disease based on a synthetic control study, and its accelerated approval of Alecensa from Roche for a specific form of lung cancer give new hope and possibilities for sustained innovation in this space.

This method of clinical trial represents a notable shift toward the use of real-world evidence (RWE). Instead of using conventional data collection methods using patients in a trial, synthetic control arms model those comparators using real-world data. The use of real-world data from sources – including electronic health records of patients, patient-generated data from medical equipment, patient registries and historical trial data – makes possible equivalent and useful comparisons.

When combined with advanced analytics, including machine learning and natural language processing (NLP), equivalent and useful trial insights become available, accelerating the pace of clinical trials while reducing costs.

Connecting with the customer – rise of digital patient engagement models

The retail world is full of examples illustrating the importance of embedding technology that fuses a consumer's physical and digital experiences. At the centre of creating such a composite experience is the ability to leverage data and the right digital channels to personalize, target and tailor a customized experience for each consumer.

For pharmaceutical brands, an urgent need to uncover the consumer's (patient's) unmet needs and effectively partner with the stakeholders in the healthcare system (clinicians, nurses, payers, patient advocates) to deliver a similar patient-at-the-centre experience is an enabler for differentiated service models.

Take, for instance, the market access programmes for patients. As the touch points between patients and the partners supporting the patient's care become digital and come through multiple sources, it results in an inconsistent, high-friction, disconnected and burdensome experience for patients engaging with a brand for their needs. The problems include repeating, confusing interactions for the patient as a result of lack of coordination among the touch points (HCP/practice, access, contact centres, insurance) and lack of a feedback loop, and the poorly understood patient journey leads to disconnects in service interactions. When one combines the scale of such operations that can typically include 500,000 or more patients on a programme – involving multiple touch points that include a 1000+ person contact centre – across brands and therapeutic areas, the problem quickly becomes very complex.

The need and opportunity for pharmaceutical brands lie in solving this problem and, in the process, gaining competitive advantage using an integrated framework that combines data science to surface new insights from service interactions. They can then go on to redesign the patient experience, effectively incorporating desirable changes in the patient's journey.

Lesson 2: The execution difference

For the life sciences business and technology executives, digital presents a distinctive set of new capabilities and differentiation opportunities for the future. But the secret to success lies in the approaches they take for execution within their organizations.

Successful digital transformations in the industry will mean a thoughtful approach and method of execution:

1. **Embracing a cloud-first strategy** – Pharmaceutical companies have until recently been slow to adopt cloud-based solutions as a mainstream strategy, often handicapped by the risks associated with the industry's regulatory implications. Unlike before, leading cloud providers, including Amazon AWS, Microsoft Azure, and Google's GCP, have built-in clinical and healthcare assets that encompass machine learning models, integrations with clinical workflows and application domain-ready data models. A cloud-first approach as central to the technology strategy enables the enterprise to rapidly develop, test and deploy production-grade applications in service of end consumer ('patient') outcomes.

2. **Automating core business processes and improving decision making with AI** – A pharmaceutical company's core business processes by design operate in an environment to increase regulatory compliance and reduce risk. However, the downside of this legacy is that it drastically retards the speed with which companies begin to realize the value of their digital transformation initiatives. By embracing a thoughtful automation strategy using robotics process automation (RPA) platforms like Ui Path, Blue Prism, or Automation Anywhere, technology executives can eliminate inefficiencies and simplify their core processes. Automating using RPA, however, is only half the answer. The other half lies in data science. By applying machine learning and NLP techniques, automated business processes become intelligent and significantly improve decision making.
 3. **Do-It-Yourself vs. Partner model** – While gone are the days for Pharmaceutical technology executives to decide between a buy vs. a build, the choice between doing it all themselves vs. selecting a Digital technology partner for execution remains. The decision framework involves addressing a few questions, which include
 - Source of differentiation: What type of intellectual property (IP) is meaningful for a pharmaceutical company to build, and how closely will it align to the primary goal of improving patient outcomes?
 - Extent of experimentation: Does the right approach lie in tackling every problem from scratch or in taking advantage of partner solutions, using accelerators that have done the hard work before? Patient data hubs, pre-configured clinical workflows, patient journey models, and pre-configured machine learning models for specific applications, to name a few.
 4. **Open Innovation and Collaboration** – Life sciences executives, particularly in R&D, leverage external partner ecosystems to accelerate their research pipeline. A similar opportunity extends to the execution of digital transformation in these organizations, where they go beyond traditional technology partner models to develop a collaborative open innovation ecosystem for joint development of new solutions at low cost.
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Nitin's conclusion:

Patient-centricity is here to stay. It is redefining the business that life sciences companies are in today and, with digital transformation in these organizations, will bring new meaning to who they serve, delivering improved patient outcomes and gaining competitive advantage in the process. Data, analytics, and a cloud-first execution strategy combined with the right choice of partnership models will place the forward-looking life sciences companies well ahead in the race to win in digital.